CPD isaac science

November 2025

[Physics Event at RGSHW 2025]

We invited 90 local schools and hosted around 170 A-level physics students, including 140 from our own school.

Sixth Form Physics

Workshop & Careers Talk

We are thrilled to announce a further event at RGS for all Physics enthusiasts! Join us as we welcome **Dr. Anton Machacek**, and **Prof Lisa Jardine-Wright** distinguished Physicists from **Cambridge University**, who have made remarkable contributions to the field of education by founding **Isaac Physics**.

Prof Jardine-Wright is the director of Isaac Physics and STEM SMART. She is fellow and Director of studies (physics) Churchill College University of Cambridge. Jardine-Wright has worked in science communication and outreach. She was a British Science Association media fellow at the *Financial Times*, and started to write for the *Times Higher Education* as well as for the journal *Science*. She has also acted as a consultant for *BBC News Magazine*.

Dr. Machacek is a Gold Medalist in the International Physics Olympiad, an alumnus of RGS, former Head of Physics and a driving force behind Isaac Physics- a platform widely used by students across the country to master challenging Physics concepts and excel in their studies.

The workshop overview:

Isaac — You work it out.

20 November 2025

Isaac Physics in High Wycombe

Programme:

10:30	Angular Motion Workshop	- see page 02
11:05	Phasors Workshop	- see page 12
11:40	Break	
12:15	Futures in Physics and University Applications to Cambridge Question and Answer session	
13:15	Lunch break	
14:15	Introduction to Special Relativity	- see page 22
15:30	End of Programme	

To see all questions set for the day, and get feedback on your answers, log into https://isaacscience.org and then enter group code 999QTB

During session

Dr Anton Machacek introduced the new concepts.

All students registered for Isaac Science and began solving problems.

Dr Machacek explained selected questions, while teachers supported their students during problem-solving.

There was also excellent networking among staff during break and lunchtime.

April 2025 Electricity and Fluids & Flights

Access to electricity booklet

https://drive.google.com/drive/folders/1HgI0PC6X8_6NKNEXn4p3ti1CzUjMUDDZ

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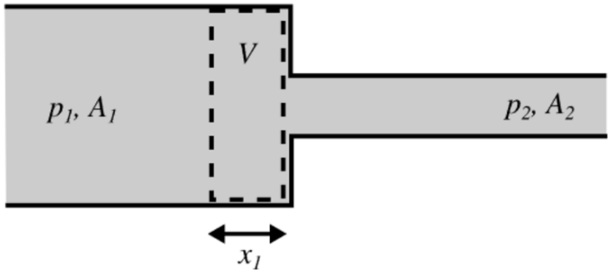
Fluids and Flight Lectures

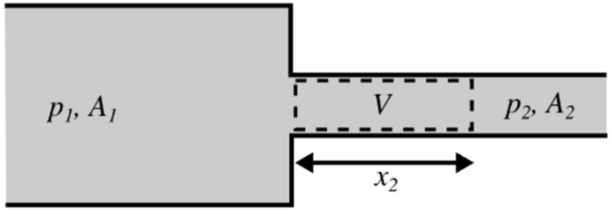
Summary of session

- The fundamental equations: continuity & energy
- Pressure at height
- Pressure and speed
- Wings
  - Lift
  - Induced Drag
  - Other drag

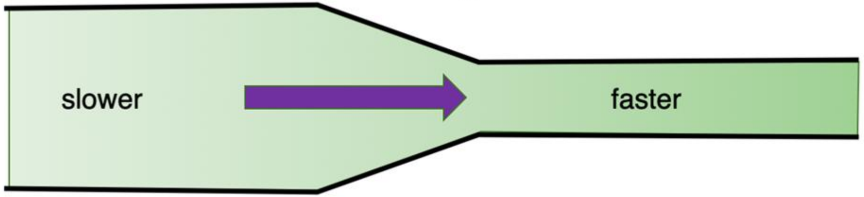
Continuity

- Incompressible fluid
- Volume flow per second
  $$\frac{V}{t} = \frac{Ax}{t} = Av$$
- $A_1 v_1 = A_2 v_2$
- Fluid speeds up when pipe narrows





How does it speed up?



- Pressure must drop.
- Work done on gas = $F_L x_L - F_R x_R = p_L A_L x_L - p_R A_R x_R = (p_L - p_R)V = -V\Delta p$
- Gain in potential energy $= mg\Delta h = \rho V g \Delta h$
- Gain in kinetic energy $= \Delta\left(\frac{1}{2}mv^2\right) = \Delta\left(\frac{1}{2}\rho V v^2\right) = \frac{1}{2}\rho V \Delta v^2$
- Conservation of energy: $-\Delta p = \rho g \Delta h + \frac{1}{2}\rho \Delta v^2$, so $p + \rho g h + \frac{1}{2}\rho v^2 = \text{const.}$